FLEET TEAM DIGEST

737MAX-FTD-34-22007

Issue Title : Unexpected Roll during LNAV operation when the RNP default changes to RNP 0.3 or less

Airplane Model	ATA	Minor Model(s)
737MAX	3461-00	-7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
737NG	9E991

Originated Date	Last Revised Date	Created On
01/17/2023	03/27/2026	01/16/2026

Estimated Completion Date	Next Update Date
01/31/2028	09/30/2026

Revision Description

Updated the following section(s):

- Milestones

Description

GE Aviation and Boeing have investigated in-service reports of unexpected roll activity resulting in lateral deviation during RNAV/ RNP approach. These events can occur on transition to RNP values of 0.3NM or less with the "B" channel Autopilot engaged in Command. When the condition occurs, the Nav Performance Scale (NPS) deviates from center, indicating that the aircraft has laterally deviated either to the left or the right from the intended path.

Dual FMC system design requirements make the left FMC primary when the FMC source select switch is set to either "Both on Left" or "Normal". Each Autopilot uses the onside FMC, meaning A/P "A" uses FMC-L for primary guidance while A/P "B" uses FMC-R. In these reported events, Autopilot "B" is selected and the FMC Source Select Switch is in the "Normal" position.

The condition appears to have been introduced with FMC software update U11, where a gain increase was added to the FMC Roll Command control law when the FMC enters the approach environment requiring RNP values of 0.3 or less. At the time the gain increases, the left FMC may contain a residual cross-track error that is not present in the right FMC. Due to the increase in command gain, the residual FMC roll command also increases and the left and right FMC command may differ by more than 5.5°. This results in a RAM Cross-load to resynchronize the two FMCs and triggers the right FMC to increase its roll command output from FMC-R and subsequent FCC "B" commanded roll rate. This generates a temporary roll maneuver that is unexpected.

GE analysis indicates a worst case deviation induced by the left FMC RAM cross-load is limited to less than 0.1 NM before the FMC self corrects. The resulting cross track error is less than the RNP containment boundary for annunciation of the "Unable RNP" message and is visible to the flight crew with display of ANP (including cross track error) deviations on the NPS.

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Boeing has determined this condition as "Non-Safety" by our Continued Operational Safety Process (COSP). Boeing has reviewed the condition related to the unexpected LNAV lateral deviation and the condition is recognizable by the flight crew and appropriate procedures are in place.

Status

This condition has been reproduced and its root cause has been determined. Boeing currently plans to incorporate the fix for this condition in FMC OPS U14.2. At this writing the incorporation schedule is still not settled.

Interim Action

There are no maintenance tasks to perform since the current software is working as implemented. Boeing does not plan on issuing any additional Flight Crew Bulletins on this workaround. Please discuss the following with your flight operations departments.

The following pilot actions can mitigate the issue:

Prior to IAF (Initial Approach Fix), de-selecting and re-selecting LNAV will clear any independent FMC roll command differences.

Use FMC-L as the primary Nav source to prevent RAM cross-loads from impacting approach procedures due to the cockpit/ avionics architecture design.

NOTE: A long direct-to may contribute to a gradual difference in roll command and the flight crew might notice FD Bar deviations differ between FMC-L and FMC-R. Automatic FMC cross-loads (triggered for any reason) prior to approach also have a mitigating effect.

Final Action

Incorporate the U14.2 service bulletin on in-service aircraft when released.

Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

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FMC OPS U14.2 - 737 MAX - Certification | No earlier than 2028 (Target)

FMC OPS U14.2 - 737 MAX - Production Incorporation | No earlier than 2028 (Target)

FMC OPS U14.2 - 737 MAX - Service Bulletin Release | No earlier than 2028 (Target)

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Fleet Team Digest Article (FTD)	737NG-FTD-34-19007		

Related Categories

- Flight Deck Effects and Maintenance Messages : Issue that occurs frequently and causes a flight deck effect or maintenance message
- Flight Operations : Flight Operations

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